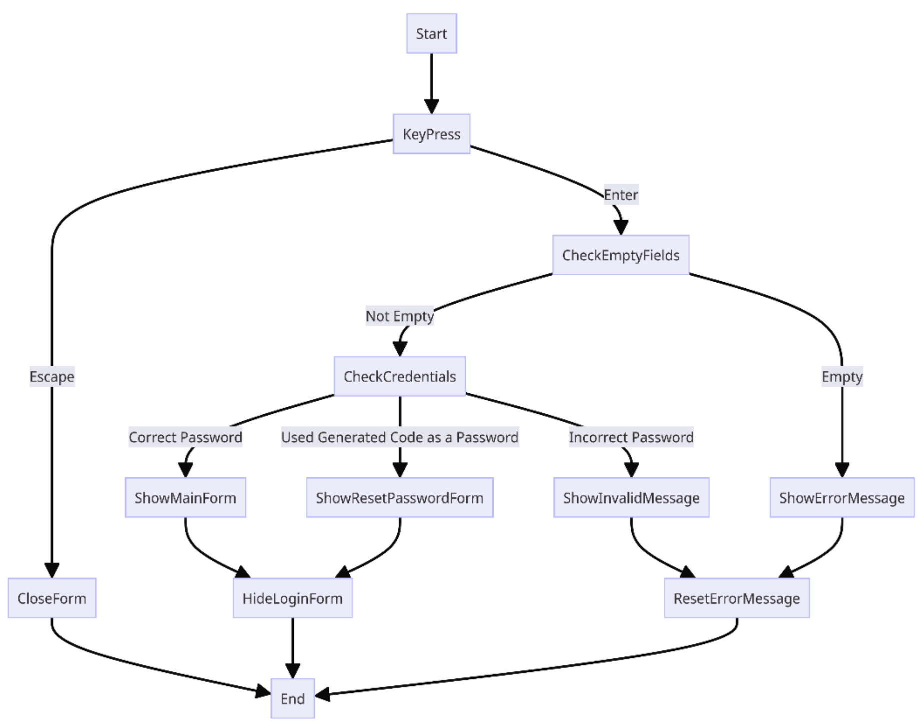
RFID Attendance System Information

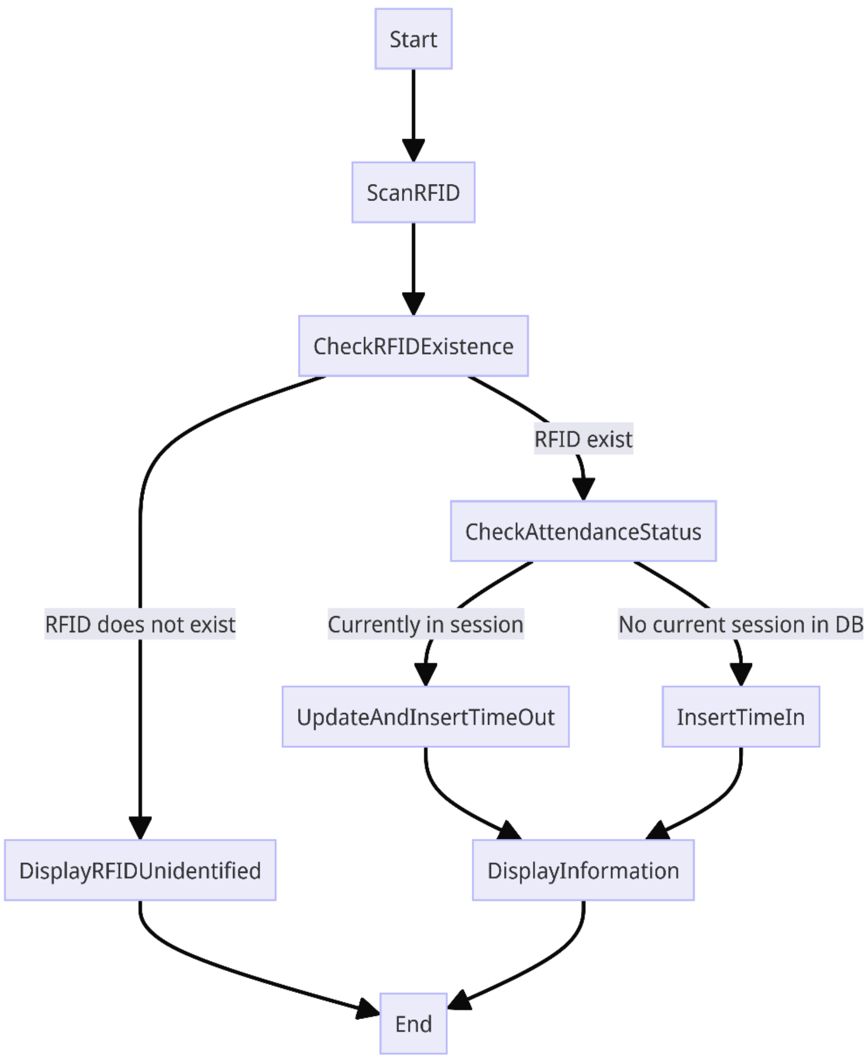
The purpose of this document is to provide more detailed flow and requirement of the RFID Attendance System. Covering its system requirements, RFID scanner Specification, Database Structure and User Manual. This document aims to give you all the details you need about the RFID Attendance System and how it's done.

Flow Chart

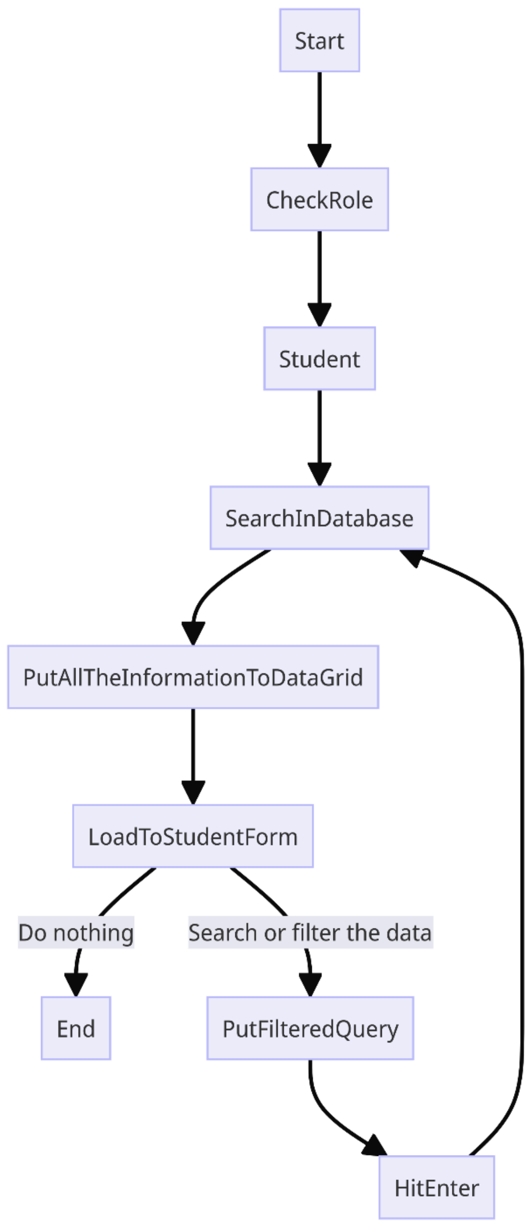
Log in flowchart - behind the scene it waits for the user to press 'Enter' to verify if the required fields are not empty. Upon confirming this, it proceeds to validate the entered credentials. If the credentials are accurate, it grants access to the main form of the program. However, if the entered password matches a generated code, it prompts the user to reset their password. In the case of an incorrect password, an error message indicating invalid credentials is displayed.



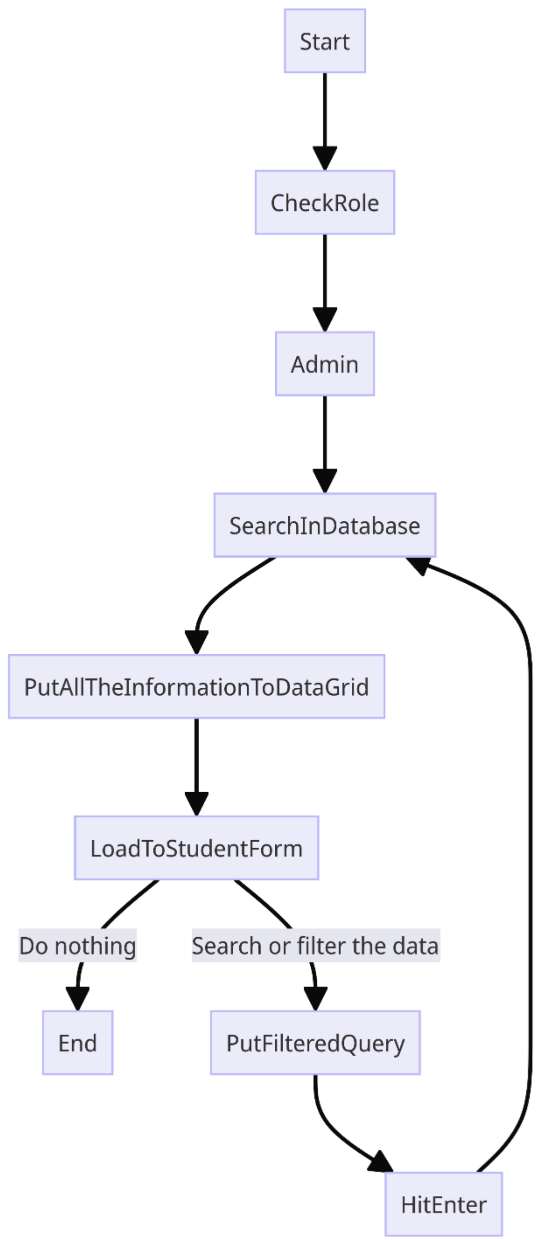
Attendance Display flowchart – It manages the insertion of attendance records. It begins by awaiting the RFID card to be tapped on the scanner. Upon detection, the system verifies if the RFID is registered. If not, it prompts a message indicating an unidentified RFID. Otherwise, it proceeds to check if the RFID is currently in a session. If it is, the system updates the data and inserts a time out. If the RFID is not currently in session, it inserts a time in to initiate a new session.



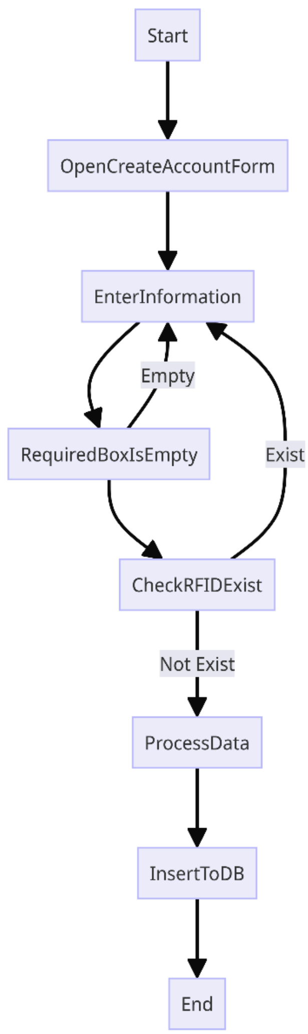
Student list flowchart – Upon clicking the student button, the system automatically retrieves all information of the registered students from the database. This includes all inputted details about each student. When the administrator filters the data using the search box, a filterQuery is initiated to search through the database and display the relevant information in the student data grid.



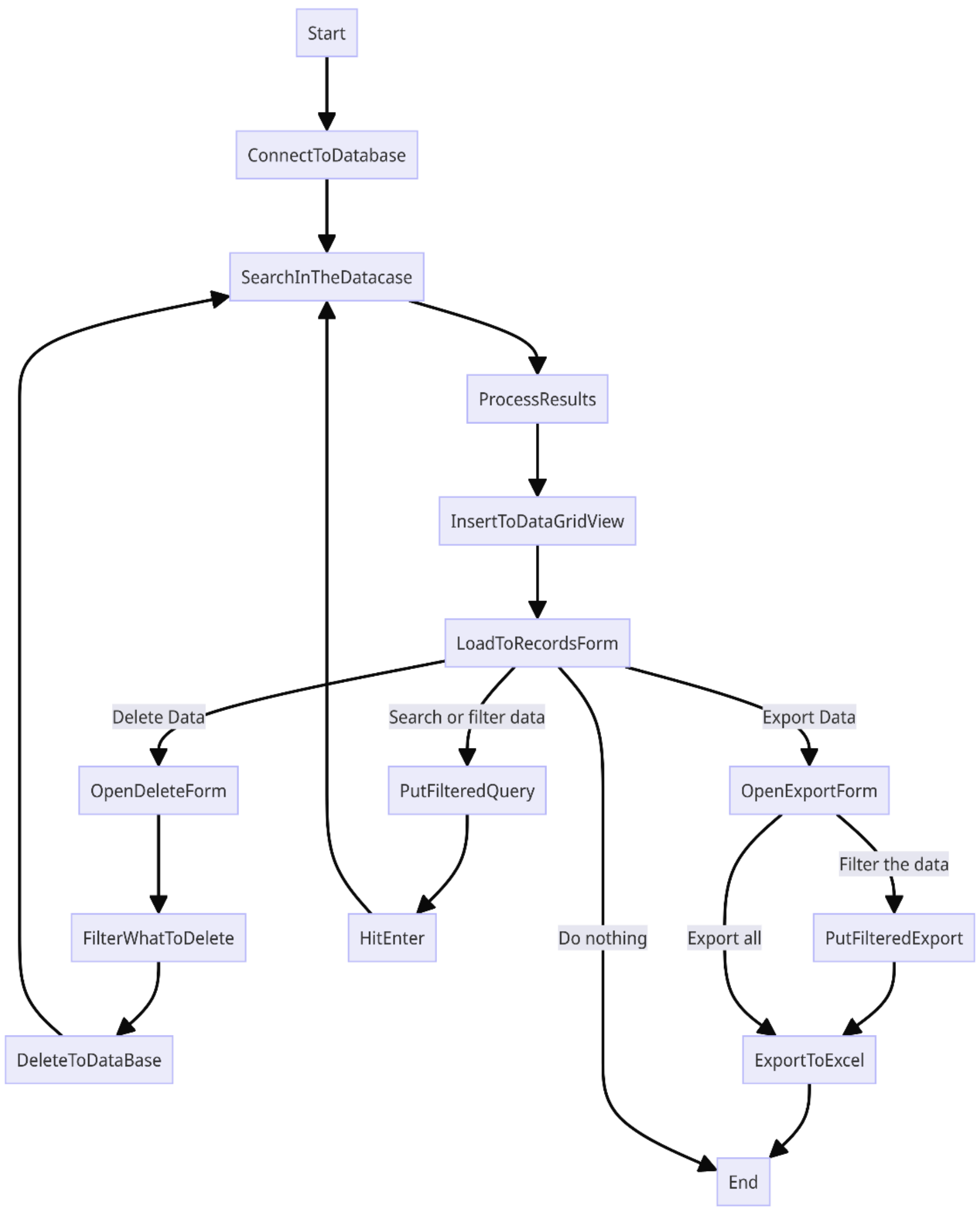
Admin list flowchart – Upon clicking the admin/teacher button, the system automatically retrieves all information of the registered admin/teacher from the database. This includes all inputted details about each admin/teacher. When the administrator filters the data using the search box, a filterQuery is initiated to search through the database and display the relevant information in the admin/teacher data grid.



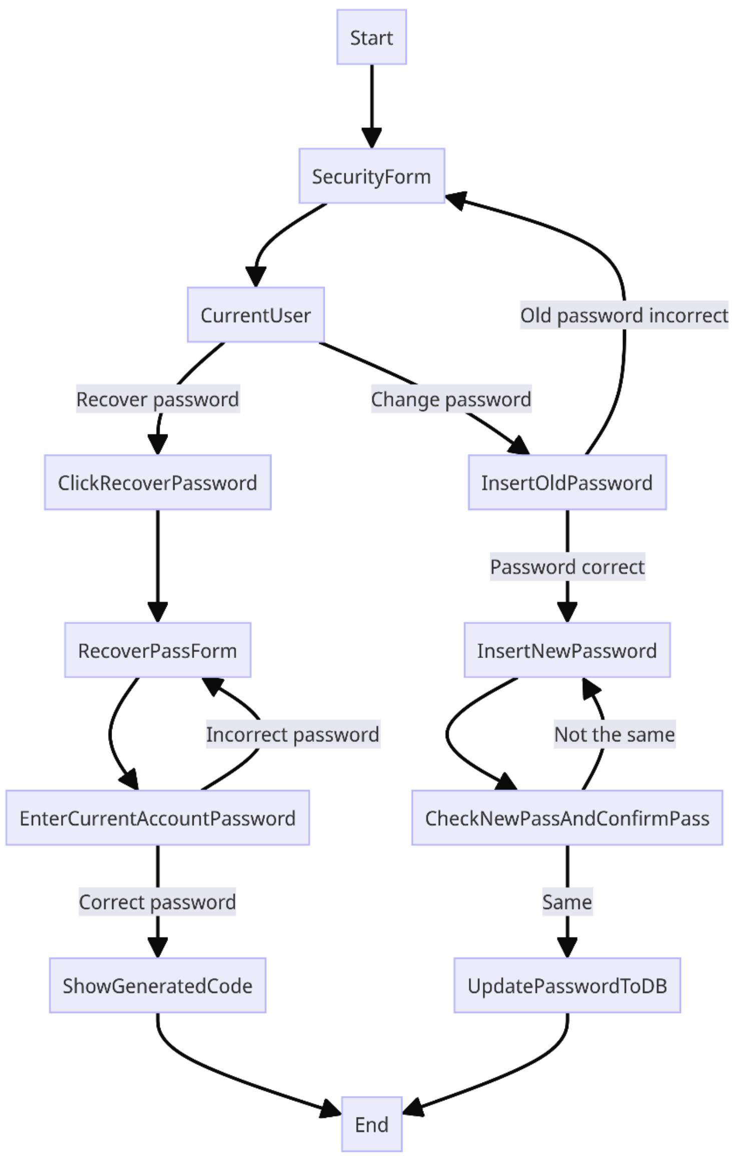
Create Account flowchart – To register an account or RFID to the system, it necessitates inputting all essential information about the individual, including RFID number, role, ID number, first name, and last name. For students, additional details such as year level and course are required. Upon submission, the system checks if all mandatory fields are filled. If they are, the process proceeds to verify if the RFID number is already registered. If not, the system inserts the information and register it into the database.



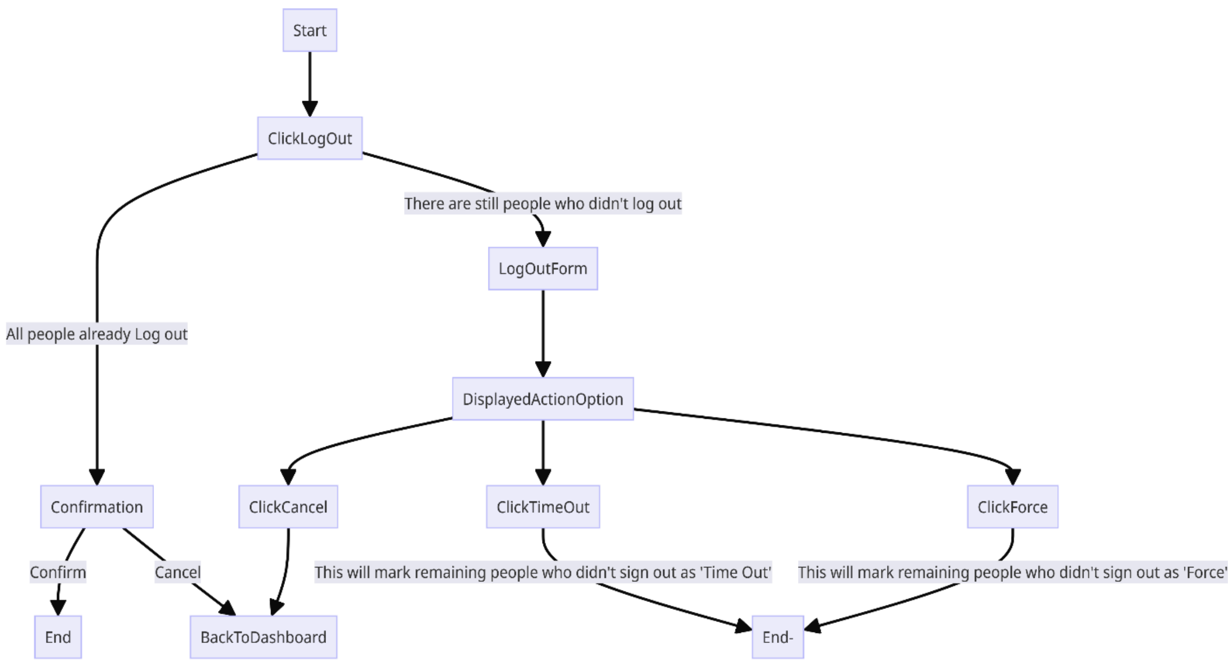
Attendance Records flowchart – Upon clicking the records button, the system automatically loads all attendance records from the database and displays them in the data grid. The administrator now has the access of all the attendance in the system. The system offers various search methods for the administrator's convenience, inserting a searchQuery to locate specific records in the database. The administrator holds the privilege to delete records from the system and also can export them to Excel file.



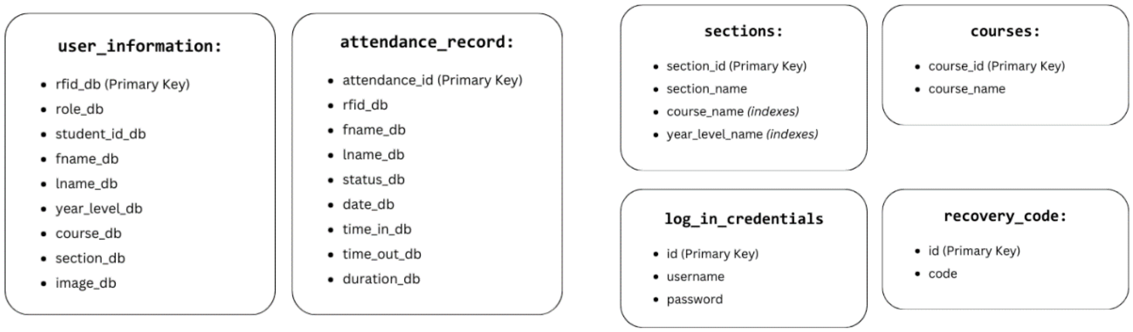
Security flowchart – Administrators has the capability to change their account passwords by providing their current password to verify their identity. Upon submission, the system verifies the accuracy of the current password. If confirmed, the administrator can proceed to update their password and synchronized it in the database. Additionally, a password recovery feature is available for instances when an administrator forgets their password. By inputting the current password of the current account, the system generates a unique code. It can use as a password in the login form to reset the account password.



Logging out flowchart – When an administrator initiates the logout process, the system doesn't immediately shut down. Instead, it conducts a check to ensure that no other users are currently active and logged in. This precautionary step is crucial for minimizing system misuse. If there are still active sessions, the administrator is presented with options: either to time out the remaining users or to forcefully log them out, indicating that the logout was not voluntary. Once the administrator selects one of these options, the system proceeds to complete the logout process and shuts down securely.



Database Structure



The database structure consists of few tables for managing different aspects of the system. The *attendance_record* table serves to track attendance, storing essential details such as RFID, first and last names, status, date, and time in/out. Courses holds info about the offered courses of the school. For logging in, there's *log_in_credentials* with usernames and passwords. If someone forgets their login, *recovery_code* helps them get back in. Sections keeps track of sections for specific course. Lastly, *user_information* captures user details, including RFID, role, student ID, names, year level, course, section, and optional images.

The database structure satisfies the requirements of the third normal form (3NF). Each table contains a primary key, fulfilling the requirements for the first normal form (1NF). Non-key attributes are fully dependent on the entire primary key, meeting the criteria for the second normal form (2NF). Also, non-key columns do not depend on other non-key columns, satisfying the requirements for the third normal form (3NF). This ensures efficient data organization and reduces redundancy, making the database structure well-optimized and following best practices for normalization.

System Requirements

Operating System: Windows 8.1 64-bit / Windows 10 Pro 64-bit

CPU: at least Intel Core i5 3rd @ 2.60Hz (4 CPUs) Gen and above

RAM: at least 4 GB and above

Hard Disk: at least 128 GB and above

Screen size: At least 5.6 inches and above

RFID Scanner Specification

Working band: 125khz Proximity Sensor

Operating system: in XP\Win CE\Win 7\Win 10\LIUNX\Vista\Android

Read card type: TK4001, EM4100, and other ID cards.

Read card distance: 0-80mm

Reading speed: < 100 mS (0.2 Sec)

Communication interface: USB

Working voltage: 5V

Working current: 100mA

Product size: (3.70 inches length * 2.36 inches width * 0.39 inches height)

Card size: 85.5 mm x 54 mm x 0.8 mm (L x W x D)

Status indicator: 2 color LED ("red" power LED, "green" status indicator)

Product weight: 33g

Package: 1 x Card Reader, 1 x 1.4m/4.5ft USB Cable, 5 x Keychain ID Cards, 5 x ID Cards